

PHY201: Waves & Optics

Problem Set 1

Aug 10, 2025

1. Given $m = 0.75 \text{ kg}$, $k = 400 \text{ N m}^{-1}$, $\omega = \sqrt{k/m} = \sqrt{400/0.75} = 23.094 \text{ s}^{-1}$. At $t = 0$: $x_0 = -4.0 \text{ cm} = -0.040 \text{ m}$ (“closer to the wall” $\Rightarrow$ negative) and $v_0 = -0.50 \text{ m s}^{-1}$ (towards the wall).

(a) Total energy

$$E = \frac{1}{2}kx_0^2 + \frac{1}{2}mv_0^2 = \frac{1}{2}(400)(0.040)^2 + \frac{1}{2}(0.75)(0.50)^2 = 0.41375 \text{ J} \approx 0.414 \text{ J}.$$

(b) Write $x(t) = A \cos(\omega t + \phi)$ with

$$A = \sqrt{x_0^2 + \left(\frac{v_0}{\omega}\right)^2} = \sqrt{(0.040)^2 + \left(\frac{0.50}{23.094}\right)^2} = 4.548 \times 10^{-2} \text{ m}.$$

Use $x_0 = A \cos \phi$, $v_0 = -A\omega \sin \phi$ to get

$$\cos \phi = \frac{x_0}{A} = -0.87944, \quad \sin \phi = \frac{-v_0}{A\omega} = 0.47601,$$

hence $\phi \simeq 2.64548 \text{ rad}$. Therefore

$$x(t) = 0.04548 \cos(23.094 t + 2.6455) \text{ m}.$$

2. The pivot is at a distance $L/3$ from the top, so the center of mass is at distance $d = L/2 - L/3 = L/6$ below the pivot. For small oscillations of a physical pendulum,

$$T = 2\pi \sqrt{\frac{I_p}{Mgd}}, \quad I_p = I_{\text{cm}} + Md^2 = \frac{1}{12}ML^2 + M\left(\frac{L}{6}\right)^2 = \frac{1}{9}ML^2.$$

Thus

$$T = 2\pi \sqrt{\frac{2L}{3g}}.$$

3. Let the unstretched length be l_0 . When hung vertically by the same mass, the extension is $\Delta l_{\parallel} = 10^{-3}l_0$ (given), and with tension T , by Hooke's law

$$Y = \frac{T/A}{\Delta l_{\parallel}/l_0},$$

$$\text{giving } YA = \frac{Tl_0}{\Delta l_{\parallel}} = \frac{mg l_0}{10^{-3}l_0} = 10^3 mg.$$

With the ends A, B held a horizontal distance l_0 apart and the mass at the midpoint, let the midpoint sag be $y \ll l_0$. Each half makes angle θ with the horizontal, $\sin \theta \simeq \tan \theta = \frac{2y}{l_0}$, and the tension T' in each half satisfies

$$2T' \sin \theta = mg \Rightarrow T' = \frac{mg l_0}{4y}.$$

The elastic stretch of each half of unstretched length $l_0/2$ is $\delta = \frac{T'}{YA} \frac{l_0}{2}$, while geometry gives the new half-length $\sqrt{(l_0/2)^2 + y^2}$. Therefore, $\delta = \frac{l_0}{2} - \sqrt{(l_0/2)^2 + y^2} \simeq \frac{y^2}{l_0}$. Equating stretches:

$$\frac{T'l_0}{2YA} = \frac{y^2}{l_0} \Rightarrow \frac{mg l_0}{4y} \cdot \frac{l_0}{2(10^3 mg)} = \frac{y^2}{l_0} \Rightarrow y^3 = \frac{l_0^3}{8000}.$$

Therefore

$$y = \frac{l_0}{20}, \quad T' = \frac{mg l_0}{4y} = 5mg \quad (\text{tension in each half of the wire}).$$

4. Look up the section "Oscillations involving massive springs" in A P French.

5. Periods depend on effective spring constant.

(a) Single spring: $T = 2\pi\sqrt{\frac{m}{k}}$.

(b) Two identical springs in parallel: $k_{\text{eff}} = 2k$, so $T = 2\pi\sqrt{\frac{m}{2k}}$.

(c) Two identical springs in series: $k_{\text{eff}} = k/2$, so $T = 2\pi\sqrt{\frac{2m}{k}}$.

6. Wire length $L = 2$ m, wire radius $a = 1$ mm = 10^{-3} m. Ultimate tensile strength $\sigma_u = 1.1 \times 10^9$ N m $^{-2}$.

(a) The largest static load before failure is

$$F_{\text{max}} = \sigma_u A = \sigma_u \pi a^2 = 1.1 \times 10^9 \times \pi (10^{-3})^2 = 3.456 \times 10^3 \text{ N}.$$

Hence the largest ball mass is

$$M_{\text{max}} = \frac{F_{\text{max}}}{g} \approx 3.52 \times 10^2 \text{ kg}.$$

If the ball is solid steel of density ρ (take $\rho \simeq 7.8 \times 10^3$ kg m $^{-3}$),

$$M_{\text{max}} = \frac{4}{3}\pi R^3 \rho \Rightarrow R = \left(\frac{3M_{\text{max}}}{4\pi\rho}\right)^{1/3} \approx 0.221 \text{ m}.$$

(b) The torsional constant for a circular wire is $\kappa = \frac{GJ}{L}$ with $J = \frac{\pi a^4}{2}$ and $G = 8 \times 10^{10}$ Pa.

For a solid sphere, $I = \frac{2}{5}MR^2$. Thus

$$T = 2\pi\sqrt{\frac{I}{\kappa}} = 2\pi\sqrt{\frac{(2/5)MR^2 L}{G(\pi a^4/2)}} = 2\pi\sqrt{\frac{4MR^2 L}{5\pi G a^4}}.$$

With the maximal M and R found above,

$$T \approx 6.57 \times 10^1 \text{ s}.$$